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Professional Profile

Bilingual specialist in embedded systems, networking, and cybersecurity with over 20 years of experience designing, securing, and optimizing complex critical infrastructures. Expert in C programming and embedded Linux systems, mastering network protocols, automation, security mechanisms, and vulnerability management. Driven by innovation and the pursuit of performance, I operate with a proactive approach focused on reliability, forward-thinking, and positive impact.

Technical Skills

- Programming Languages: C, C++, Python, PhP, Perl, Java, C#
- Embedded Development: STM32, CubeMX, CubeIDE, TouchGFX, RTOS (Azure ThreadX and FreeRTOS), PKA, SAES, TZ (ARM TrustZone), TEE
- Networking: IP, BGP, MPLS, OSPF, VXLAN, VPN, SASE, IOS (including XE and XR), NXOS
- Monitoring: MRTG, RRDTOOL and SNMP
- Operating Systems: Linux, Windows (server and workstation), Solaris
- Security: Snort, Suricata, Trend Micro EDR/NDR/DDAN/DDI, Checkpoint FW, Cisco ASA/FTD, SIEM (IBM, AlienVault, Prelude), IDS/IPS/NDR, auditing, hardening, incident response
- Virtualization and Cloud: VMware, VirtualBox, Docker, Vagrant, vCloud Director, AWS, Azure
- Automation & DevOps Tools: Ansible, Git, PyATS Genie, Azure DevOps (ADO), Atlassian

Professional Experience

Infrastructure Consultant – Telecommunications

Société d'Habitation du Québec (SHQ) – Québec, QC (Hybrid)
July 2024 – February 2025

As a specialist in telecommunications infrastructure, I designed and enhanced critical architectures integrating advanced technologies, directly contributing to the security, performance, and operational autonomy of SHQ's network.

Key Impacts:

- Strengthened network security through the deployment of Checkpoint Harmony SASE, enhancing VPN access protection and employee web browsing.
- Executed a planned and documented migration of inter-governmental links (RGT), minimizing operational impact through a controlled interruption.
- Accelerated diagnostics and network interventions via the development of an internal Python-based automation tool (PyATS/Genie).
- Improved compliance and network resilience by implementing a structured and standardized inventory of network assets.
- Reduced incident risk by proactively replacing outdated firewalls with secure, next-generation equipment.
- Enhanced traceability and auditability by standardizing SYSLOG and network access logs, facilitating anomaly detection and regulatory compliance.

Additional Contribution: Delivered comprehensive technical documentation, configuration and operation guides to internal teams, ensuring a smooth transition and sustainable management of the implemented solutions.

Cyberdefense Consultant

Ministry of Cybersecurity and Digital Affairs of Québec (MCN) – Québec, QC (Hybrid)
July 2023 – January 2024

Assigned to the Cyber Defense Operations Center (DCOCD), I contributed to the protection of Québec's critical digital assets by ensuring active threat monitoring, rapid incident response, and continuous improvement of the defense environment.

Key Impacts:

- Reinforced overall security posture by ensuring availability and performance of detection tools, including EDR systems, antivirus, and secure gateways (Service Gateway).
- Improved operational efficiency through remote interventions via EDRs, reducing remediation time with no service interruptions.
- Validated system compliance within the Trend Vision One ecosystem, ensuring consistent reporting and alerting across both client and cloud environments.
- Supported service resilience by proactively monitoring critical AWS-hosted servers, aiding in early anomaly detection.

Quality Assurance Technician

Cyberkar Systems – Québec, QC (Remote)
May 2021 – April 2022

Responsible for the quality assurance of embedded technology products (Kommander) used in emergency response vehicles. I ensured the functional, software, and documentation compliance of critical solutions for first responders.

Key Impacts:

- Improved the reliability of embedded systems by thoroughly documenting new software features and validating their behavior through structured testing.
- Optimized issue tracking and deliverable management using the Jira platform, ensuring clear and traceable deployment of fixes.
- Standardized user and technical documentation using Confluence, streamlining feature integration and user training.
- Ensured product quality by applying functional test plans in X-Ray, guaranteeing appropriate coverage before deployment in critical environments.

Senior IP Network Analyst

Vidéotron Ltée – Montréal & Québec, QC (Hybrid)
July 2018 – October 2020

Responsible for validating, deploying, and maintaining critical IP interconnections for telephony, voicemail, and mobility services. I ensured service reliability while modernizing the network infrastructure.

Key Impacts:

- Ensured reliable and resilient network connectivity by implementing advanced configurations (VXLAN, MPLS-BGP, L2VPN) for strategic IP services.
- Improved mobile service performance by migrating wireless roaming links to 10 Gb capacity, reducing latency.
- Optimized traffic flows for FIZZ, contributing to a nomination for an internal excellence award.
- Strengthened interconnect documentation, simplifying incident resolution and knowledge transfer for junior IP analysts.
- Coordinated critical technical projects while acting as a mentor and point of contact between various internal departments.

Professional Services Engineer

Trend Micro Canada – Québec, QC (Remote)
November 2011 – March 2017

Delivered professional cybersecurity services to clients and partners across Canada. Supported deployments, trained users, simulated attacks, and demonstrated solution value in production and lab environments.

Key Impacts:

- Accelerated the adoption of security solutions through bilingual training tailored to real-world environments.
- Reinforced security postures by customizing and optimizing Trend Micro product deployments based on the constraints of critical infrastructures.
- Prevented major incidents via penetration tests and data exfiltration simulations to validate system robustness.
- Shortened project delivery timelines by setting up a complete simulation lab for faster evaluation and demonstrations.
- Boosted client retention by maintaining close technical expertise and providing personalized support.

Network Security Analyst & Information Security Advisor

Victrix – Québec, QC (Hybrid)
June 2010 – June 2011

Security specialist assigned to technology projects for government and municipal organizations. I deployed, analyzed, and optimized cybersecurity solutions while serving as a key technical representative for partner vendors.

Key Impacts:

- Enhanced threat visibility by integrating advanced detection technologies tailored to sensitive environments.
- Strengthened technical partnerships by serving as the primary contact between Victrix and Trend Micro, ensuring solution alignment.
- Reduced intrusion risks through in-depth analysis of network communications and finely tuned security mechanisms.
- Accelerated incident diagnostics by installing antivirus and SIEM systems (Trend Micro, McAfee, OSSIM) configured for client-specific environments.

IT and Information Security Advisor

BPR – Québec, QC (Hybrid)
March 2010 – June 2010

Mandated to strengthen the protection of the organization's informational assets, both technologically and procedurally. I played a strategic role in advising and intervening to optimize security practices.

Key Impacts:

- Enhanced endpoint and server security by deploying antivirus solutions (Trend Micro) and implementing data protection controls.
- Reinforced IT governance by recommending precise adjustments to the onboarding process, resulting in better access management.
- Contributed to organizational compliance by aligning security practices with business standards and operational needs.

IT Security Analyst & Advisor

Nurun – Québec, QC (Hybrid)
August 2009 – March 2010

Responsible for security in major government mandates. I conducted audits, vulnerability testing, and risk analyses to support governance and digital resilience for critical environments.

Key Impacts:

- Strengthened the security of the Photo Radar project (MTQ) by validating infrastructure and managing sensitive data.

- Improved asset classification by developing a custom web application to streamline risk analysis.
- Reduced critical vulnerabilities by conducting OWASP Top 10 tests and delivering actionable technical reports.
- Supported large-scale projects like Québec's Health Record (DSQ) by contributing to security architecture at the Ministry of Health (MSSS).
- Optimized incident handling at the CNESST by revising internal processes.
- Influenced strategic decisions by drafting security advisories aligned with ministry needs.

IP Systems and Network Security Analyst

Société de Gestion Informatique du Québec (CSO – MSSS DMZ) – Québec, QC (Hybrid)
June 2007 – February 2009

Assigned to the demilitarized zone of Québec's Health Network, I played a key role in 24/7 operational cybersecurity, ensuring continuity and evolution of critical systems protecting sensitive data.

Key Impacts:

- Increased proactive threat detection through strategic Snort probe deployments feeding SIEM systems (Prelude and Trigeo).
- Reinforced security posture with centralized antivirus, IDPS, web filtering, and encryption systems for sensitive endpoints.
- Improved real-time network visibility by building custom performance dashboards using RRDTool/MRTG.
- Secured strategic communications by managing security farms for web (Trend Micro) and email (Barracuda) traffic.

Senior IP Network Security Analyst

Bell Canada – Connexim – Montréal, QC
November 2005 – May 2007

In charge of secured network environments for major institutional clients, including Mouvement Desjardins. My Level 3 technical role supported strategic units (Level 4), helping to maintain strong cybersecurity and supporting infrastructure evolution.

Key Impacts:

- Maintained high availability of critical services through administration of VPNs, firewalls, SIEMs, and multi-tenant IDPS systems.

- Reduced incident risks with efficient Level 3 intervention management during network attacks or service outages.
- Contributed to secure infrastructure evolution by participating in updates and interconnection projects for sensitive networks.
- Improved threat response by coordinating escalations with the Security Operations Center (CGSR).

IP Network Security Analyst

Bell Canada – Connexim – Montréal, QC
November 2004 – November 2005

Assigned to the Network Management Center (CGSR), I was responsible for security operations of critical network environments for clients such as Desjardins, Alcan, and Loto-Québec. I ensured infrastructure stability and protection across Québec.

Key Impacts:

- Strengthened intrusion detection through proactive IDS console monitoring and analytical reporting.
- Reduced incident response time by providing Tier 1–3 support for client infrastructure.
- Secured network changes via deep analysis and rigorous firewall/IDPS configuration.
- Improved client security posture by working closely with internal teams and partner security groups.

Network System and IP Security Administrator

Vidéotron Télécom Ltée – Montréal, QC (Hybrid)
October 1998 – March 2004

Key member of the Internet engineering and core network team, responsible for the development, maintenance, and monitoring of security and interconnection systems linking Vidéotron with its North American partners.

Key Impacts:

- Continuously improved network performance by deploying communication analysis and traffic evaluation tools.
- Continuously maintain and developed monitoring applications with MRTG, RRDTOOL & SNMP.
- Optimized access management through administration of critical systems for security, billing, and monitoring across access and backbone networks.

- Modernized the billing architecture by developing a NETFLOW-based system ensuring reliable, traceable revenue data.
- Centralized provincial network management by implementing HP OpenView NNM and standardizing operating procedures.
- Ensured operational continuity by supporting Sun, Linux, Cisco, and PIX firewall environments, notably during Y2K transition.

Intern – Programmer Analyst (Final Year Internship)

Vidéotron Télécom Ltée – Montréal, QC
May 1998 – October 1998

Integrated into the video department in charge of broadcasting for major broadcasters (TVA, Québecor), I supported engineering in evaluating digital TV technology solutions.

Key Impacts:

- Improved critical equipment management by designing and programming a video encoder supervision app using the TL1 protocol.
- Strengthened infrastructure resilience by performing full hardware validation to ensure Y2K compliance.
- Contributed to the company's tech evolution by supporting the integration of new digital broadcasting standards.

Education

Bachelor's Degree Computer Science – Laval University (1998)

Certifications or corporate training

- CISSP – Certified Information Systems Security Professional (2009)
- CEH – Certified Ethical Hacker (2008)
- CCNA – Cisco Certified Network Associate (2004)
- ITIL Foundations – Information Technology Infrastructure Library (2007)
- AWS Cloud Practitioner & Well-Architected (2017)
- Ansible & Network Programming – IBM / Global Knowledge (2019–2020)
- Additional: Checkpoint, HP NNM, Incident Response, ETS Project Management (1999–2013)

Languages

- French: Native
- English: Professional Proficiency